

CAMPUSRSSI: Real-World Wi-Fi RSSI Measurements for Indoor Localization

JinYi Yoon*, Yeawon You†, Dayeon Kang‡, Jeewoon Kim§, and HyungJune Lee¶

*Department of Computer Science, Virginia Tech, Blacksburg, VA, USA

†Department of Cyber Security, Ewha Womans University, Seoul, South Korea

‡Manning College of Information & Computer Sciences, University of Massachusetts, Amherst, MA, USA

§Department of Computer Science, University of California, Irvine, CA, USA

¶Department of Computer Science and Engineering, Ewha Womans University, Seoul, South Korea

Email: hyungjune.lee@ewha.ac.kr

Abstract—Received Signal Strength Indicator (RSSI) values from Access Points (APs) are essential in fingerprint-based indoor localization. However, there are few datasets of dense and precise data points due to the labor-intensive collection. We provide CAMPUSRSSI, a dataset of real-world Wi-Fi RSSI measurements collected in university campus buildings. We collect RSSI measurements 1) with every 30 cm granularity; 2) 158 measurements at each reference point on average; 3) under four differently obstructed in-building places; and 4) with the x-y coordinates of APs, floorplans, and obstacle information. The dataset¹ is available at <https://github.com/jinyi-yoon/CampusRSSI.git>.

I. INTRODUCTION

Indoor localization determines a user's position in indoor environments, where satellite-based positioning such as Global Positioning System (GPS) is inapplicable. One of the most easily applicable localization systems is device-based radio fingerprinting, where a user holds a Wi-Fi-enabled device and estimates its position using the measured signal values from the device. Although the Received Signal Strength Indicator (RSSI) vector map is essential for radio fingerprint-based localization, there are few datasets with RSSI values with the coordinates of reference points and floor plans. Moreover, since it is very labor-intensive to collect real-world data with exact coordinates, it is hard to collect fine-grained data.

We present datasets with radio fingerprint maps consisting of Wi-Fi RSSI measurements for device-based indoor localization. We collect the data in four different indoor environments. We deployed access points (APs) in each place and measure the RSSI values using a customized Android application. The main contribution of this CAMPUSRSSI dataset can be summarized as follows:

- We offer fine-grained RSSI measurements with every 30 cm granularity with their corresponding coordinates.
- As the RSSI dynamically varies, we provide 158 measurements on average for each coordinate.
- We measure the RSSI of in-building places in four different levels of obstructions.
- We provide the x-y coordinates of APs, floorplans, and obstacles with geographical maps and signal heatmaps.

¹This dataset has been originated from *GAN-Loc* [1]. If you are interested in using CAMPUSRSSI dataset, please cite *GAN-Loc* [1] and *CollageMap* [2].

II. DATA COLLECTION

We collect data from differently structured places (Fig. 1):

- *Free-Obs (Hall)*: 10 APs over $9.6 \times 9.9 m^2$, where the place is relatively open with a single round pillar
- *Low-Obs (Lounge)*: 11 APs over $6.6 \times 9.9 m^2$, where the area is partitioned by the waist-high wooden wall
- *Medium-Obs (Office A)*: 16 APs over $9.9 \times 9.9 m^2$, where the room with AP #6 and #15 is enclosed by concrete walls, other rooms are enclosed by glass walls, and the black-colored boxes are the concrete pillars, while all of the doors are opened
- *High-Obs (Office B)*: 10 APs over $10.8 \times 12.0 m^2$, where the concrete wall at $x = 8.4 m$ divides the office area and the hall where the AP #9 is located, and all of the rooms are enclosed by glass walls, while the rooms with AP #0, #2, and #4 are closed, and the rooms with AP #1, #3, and #5 are opened

Methodology. We deploy wireless APs of EFM-Networks ipTIME N104Q-i with a single antenna of dBi. It communicates using IEEE 802.11n of 150 Mbps over 2.4 GHz band. We place the APs on the ground except for AP #12 and #14 in *Medium-Obs (Office A)*. To collect the fine-grained data with every 30 cm, we installed tiles with the size of $0.3 \times 0.3 m^2$ on the floor. We collect the data using an Android application for RSSI data collection on 26 LG G Pad 3 10.1 (LG-X760) equipped with Wi-Fi. We keep the position of devices at the left bottom of the tile. To record realistic and dynamic RSSI logs, we did not control any movements of pedestrians.

We develop our own Android application to periodically record RSSI values by using `wifiManager`. Due to the inherent latency issue of the `wifiManager`, we have measured the RSSI value of communicable APs with an interval of three seconds to avoid the same RSSI measurements.

III. DATASET DESCRIPTION

Using a customized application for RSSI collection, we have recorded the RSSI values from all communicable APs. Since there are some unknown APs and null values, we have preprocessed the raw data and merged the multiple files from different Android devices into a single .csv file

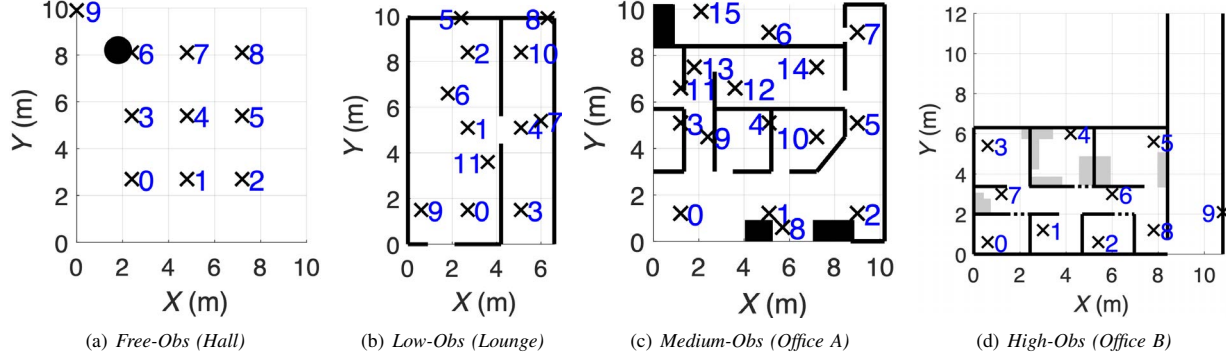


Fig. 1. The four different places, where the location of AP is marked with 'x' along the AP ID. The black line and boxes are permanent obstacles such as the walls and pillars, and the gray boxes are the movable obstacles such as bookshelves.

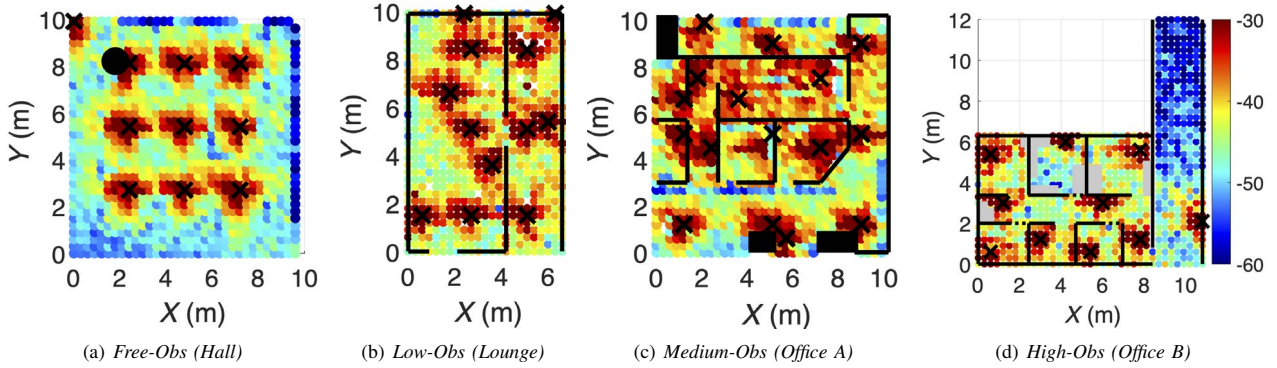


Fig. 2. The heatmap of the maximum RSSI value in each point.

TABLE I
AN EXAMPLE OF 'NODELOC_SS.CSV' FILE

Timestamp	X (m)	Y (m)	AP0	AP1	AP2	AP3	AP4
97	0	0	-56	-58	-65	-52	-53
571	0	0	-55	-58	-67	-54	-49
640	0	0.3	-56	-60	-67	-53	-49
811	0	0.3	-56	-58	-60	-53	-55
967	0	0.6	-56	-56	-60	-52	-53

for each place. First, we have only left our own dedicated APs and removed APs with unknown locations or different specifications. Second, we have removed the points outside the Region of Interest (RoI). Third, we treat consecutive identical RSSI vectors recorded multiple times in sequence as a single measurement, eliminating duplicates. Fourth, we have replaced the null value with the recent RSSI. Lastly, we have merged multiple preprocessed files into a single file for each place.

The refined dataset consists of two files for each place: 1) 'aploc.csv': the location of each AP; and 2) 'nodeloc_ss.csv': the measured RSSI value sets for each reference point.

AP location file. A file 'aploc.csv' for each place records the x-y coordinates of APs. The row ID is corresponding to the AP ID (starting from 0), and the first and the second columns denote the x and y coordinates of each AP, respectively.

RSSI file. A file 'nodeloc_ss.csv' consists of three main values: 1) timestamp: the first, second, third, and fourth columns denote the elapsed time in hours, minutes, seconds,

and milliseconds, respectively; 2) coordinate: the fifth and sixth columns denote the coordinate of X and Y in meters; and 3) measured RSSI: the measured value from each connectable AP in following columns. We illustrate the heatmap of RSSI values for each place in Fig. 2. We apply a specific color on each coordinate with the maximum RSSI value among all APs.

IV. CITATION

CAMPUSRSSI is available at <https://github.com/jinyi-yoon/CampusRSSI.git>. This dataset has been originated from *GAN-Loc* [1]. If you are interested in using this dataset, please cite *GAN-Loc* [1] and *CollageMap* [2].

ACKNOWLEDGMENTS

This work was supported by the Institute of Information & communications Technology Planning and Evaluation (IITP) grant funded by the Korea government (MSIT) (No. RS-2022-00155966, Artificial Intelligence Convergence Innovation Human Resources Development (Ewha Womans University)).

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